EHDA PORTAL

SPRINT 1 AND SPRINT 4 WORK REPORT

Online Blood Donation Platform

Prepared by: Samira Azizi

Role: Developer 1 — Authentication and Donor Account Foundation

Technology: Python, Django, HTML, CSS, and SQLite/Django database tools

Sprint 1: October 10–14, 2026

Sprint 4: October 24–29, 2026

1. Overview

This paper analyzes Ehda Portal in terms of its account creation structure, authentication process, donor profiles and associated account functionalities. The Sprint 1 covers user registration, authentication (login/logout), security for authentication, donor profile, blood type, and availability. The Sprint 4 focuses on the donor's donation history and management of account details.

The implementation uses Django and Python as back-end technology, Django forms for data entering by the user and validation and the project's database for account details storage.

2. Sprint 1: Foundation

2.1 Requirements Covered

Backlog ID	Task	What Was Delivered
#021	Project Structure and Database Connection	Django project organization and database integration for account information.
#001	User Registration	Registration form for account creation, including personal details, email, phone, number, password, and role.
#002	Login and Logout	Authentication using email and password, with session management and logout functionality.
#022	Authentication Security and Permission	Password protection and access restrictions for authenticated users.
#003	Create and Update Donor Profiles	Donor profile fields and forms for viewing and updating personal information.
#004	Blood Type and Availability Fields	Blood type and availability fields within donor information
#005	Change Availability Status	Functionality for donors to indicate their availability

#024	Input Validation and Error Handling	Form validation for required information, duplicate email addresses, and password confirmation.
#025	Initial Test Cases and Defect Tracking	Test scenarios covering registration, authentication, donor profiles, validation, and account access.

2.2 Implementation Summary

The authentication framework supports the life cycle of the user account, from its creation till login and logout. During the registration, the system collects all necessary account details and performs their validation. The passwords are stored using Django's password hashing technique.

Donor's profile management involves the personal details of users as well as their blood type and availability. The authentication and access control is used to protect the account-related pages. The form validation provides feedback in case of entering invalid data or registering with existing email account.

2.3 Testing

The Sprint 1 test scenarios cover:

- Registration with valid information and validation of duplicate email addresses.
- Password confirmation and password security.
- Login with valid and invalid credentials.
- Logout and access restrictions for protected pages.
- Donor profile updates.
- Blood type and availability information.
- Input validation and error messages.

2.4 Sprint 1 Diagrams and Evidence

Sprint 1 has included the use case and sequence diagrams for the features: Register, Login/Logout, Donor Profile, Blood Type, Availability, Authentication. Evidence includes recorded videos of the application, examples of validation, Output of Django checks and commits/Pull requests on Github.

3. Sprint 4 — Donation History and Account Management

3.1 Requirements Covered

Backlog ID	Task	What Was Delivered
#015	Donation History	A user-facing feature for viewing donation records associated with the user's account.
#016	Manage Account Information	Account information management for updating permitted personal details.

3.2 Summary of Implementation

#015 Donation History

This allows the user to view their donation history. It will list out all donations made by the user that have been authenticated. If there are no donations it will handle this.

#016 Manage Account Details

This allows the user to manage their personal account details. This can include fields such as first name, last name, email and phone number. Validation helps keep the data accurate and stops duplicate emails. Users can only edit their own information that they are allowed to.

3.3 Testing

The Sprint 4 test scenarios cover:

- Access to donation history through an authenticated account.
- Display of the appropriate user's donation records.
- The donation history page when no records exist.
- Updating account information with valid input.
- Validation of invalid email addresses and duplicate email addresses.
- Saving and displaying updated account details.
- Restricting account changes to the appropriate user and permitted fields.

3.4 Sprint 4 Diagrams and Evidence

The Sprint 4 includes a use case diagram for view my donations and manage account, as well as sequence diagrams to show how the user will request, authenticate, verify ownership, get data from database and receive a response.

Evidence could include recorded video of the view my donations page, account form, validation messages and test outputs.

4. Tools Used in Development

Python and Django has been used to develop the backend, with Django forms for handling input. Git has been used with GitHub for version control.

GitHub link: https://github.com/Kaihansaba/Ehda-Portal

This is where the source code for the project can be found. It has helped with collaboration with my development team.

5. Conclusion

The task I have been given for Ehda Portal involved creating the login and account features for donors in Sprint 1. For Sprint 4 I created the view donations and manage account features. This allows users to securely access their account and have easy access to their personal account information and donations.

This task has contributed to the overall platform by providing the account functions required by donors and allowing integration with the features created by the rest of the team.